

Analytical Solution of AC-Driven RLC Circuit

The second-order differential equation of current for an AC-driven RLC circuit is,

$$I'' + \frac{R}{L}I' + \frac{1}{LC}I = \frac{V_0}{L}\omega \cos \omega t$$

$$\text{let } \alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}, \quad F_0 = \frac{V_0}{L}\omega$$

$$\boxed{I'' + 2\alpha I' + \omega_0^2 I = F_0 \cos \omega t} \quad (1)$$

Complimentary Solution :

$$\text{let } I'' + 2\alpha I' + \omega_0^2 I = 0$$

$$I = e^{mx}, \quad I' = me^{mx}, \quad I'' = m^2 e^{mx}$$

$$m^2 e^{mx} + 2\alpha(me^{mx}) + \omega_0^2 e^{mx} = 0$$

$$m^2 + 2\alpha m + \omega_0^2 = 0$$

$$m_{1,2} = \frac{-2\alpha \pm \sqrt{(2\alpha)^2 - 4(1)(\omega_0^2)}}{2(1)}$$

$$= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

We are interested in underdamped cases ($\omega_0 > \alpha$)

$$m_{1,2} = -\alpha \pm \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha \pm i\sqrt{\omega_0^2 - \alpha^2} = -\alpha \pm i\omega_d$$

$$\text{where } \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

The roots are complex, therefore the complimentary solution is given as

$$I_c = e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

Particular Solution :

$$F_0 \cos(\omega t) \Rightarrow I_p = B_1 \cos \omega t + B_2 \sin \omega t$$

$$I'_p = -B_1 \omega \sin \omega t + B_2 \omega \cos \omega t$$

$$I''_p = -B_1 \omega^2 \cos \omega t - B_2 \omega^2 \sin \omega t$$

Substitute into the differential equation (1)

$$-B_1 \omega^2 \cos \omega t - B_2 \omega^2 \sin \omega t + 2\alpha(-B_1 \omega \sin \omega t + B_2 \omega \cos \omega t) + \omega_0^2(B_1 \cos \omega t + B_2 \sin \omega t) = F_0 \cos \omega t$$

comparing coefficients of $\cos \omega t$

$$-B_1 \omega^2 + 2\alpha \omega B_2 + \omega_0^2 B_1 = F_0$$

$$B_1(\omega_0^2 - \omega^2) + 2\alpha \omega B_2 = F_0 \quad (2)$$

comparing coefficients of $\sin \omega t$

$$-B_2 \omega^2 - 2\alpha \omega B_1 + \omega_0^2 B_2 = 0$$

$$B_2(\omega_0^2 - \omega^2) - 2\alpha \omega B_1 = 0$$

$$B_1 = \frac{B_2(\omega_0^2 - \omega^2)}{2\alpha \omega} \quad (3)$$

substitute (3) into (2)

$$\begin{aligned}\left(\frac{B_2(\omega_0^2 - \omega^2)}{2\alpha\omega}\right)(\omega_0^2 - \omega^2) + 2\alpha\omega B_2 &= F_0 \\ B_2 \left(\frac{(\omega_0^2 - \omega^2)^2}{2\alpha\omega} + 2\alpha\omega\right) &= F_0 \\ B_2 \left(\frac{(\omega_0^2 - \omega^2)^2 + (2\alpha\omega)^2}{2\alpha\omega}\right) &= F_0 \\ B_2 &= \frac{F_0(2\alpha\omega)}{(\omega_0^2 - \omega^2)^2 + (2\alpha\omega)^2}\end{aligned}$$

let $D = (\omega_0^2 - \omega^2)^2 + (2\alpha\omega)^2$

$$\boxed{B_2 = \frac{F_0(2\alpha\omega)}{D}} \quad (4)$$

Substitute (4) into (3)

$$\begin{aligned}B_1 &= \frac{\left(\frac{F_0(2\alpha\omega)}{D}\right)(\omega_0^2 - \omega^2)}{2\alpha\omega} \\ \boxed{B_1 &= \frac{F_0(\omega_0^2 - \omega^2)}{D}}\end{aligned} \quad (5)$$

↪ Particular Solution

$$\therefore I_p = B_1 \cos \omega t + B_2 \sin \omega t$$

where

$$\begin{aligned}D_1 &= (\omega_0^2 - \omega^2)^2 - (2\alpha\omega)^2 \\ B_1 &= \frac{F_0(\omega_0^2 - \omega^2)}{D_1} \\ B_2 &= \frac{F_0(2\alpha\omega)}{D_1}\end{aligned}$$

The Full solution of equation (1) is given by

$$\begin{aligned}I &= I_c + I_p \\ &= e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t) + B_1 \cos \omega t + B_2 \sin \omega t\end{aligned}$$

Initial Values : $I(0) = 0$

$$\begin{aligned}0 &= A_1 + B_1 \\ A_1 &= -B_1\end{aligned}$$

$$I' = -\alpha e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t) + e^{-\alpha t}(-A_1 \omega_d \sin \omega_d t + A_2 \omega_d \cos \omega_d t) - \omega B_1 \sin \omega t + B_2 \omega \cos \omega t$$

Initial Values : $I'(0) = 0$

$$\begin{aligned}0 &= -\alpha A_1 + A_2 \omega_d + \omega B_2 \\ &= \alpha B_1 + A_2 \omega_d + \omega B_2 \\ A_2 &= \frac{-\alpha B_1 - \omega B_2}{\omega_d}\end{aligned}$$

✓ Equation of current (I)

$$\therefore I = e^{-\alpha t} \left[-B_1 \cos \omega_d t - \left(\frac{\alpha B_1 + \omega B_2}{\omega_d} \right) \sin \omega_d t \right] + B_1 \cos \omega t + B_2 \sin \omega t$$

where

$$D_1 = (\omega_0^2 - \omega^2)^2 - (2\alpha\omega)^2$$

$$B_1 = \frac{F_0(\omega_0^2 - \omega^2)}{D_1}$$

$$B_2 = \frac{F_0(2\alpha\omega)}{D_1}$$

The second-order differential equation of current for an AC-driven RLC circuit is,

$$Q'' + \frac{R}{L}Q' + \frac{1}{LC}Q = \frac{V_0}{L}\sin \omega t$$

$$\text{let } \alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}, \quad F_0 = \frac{V_0}{L}\omega$$

$$Q'' + 2\alpha Q' + \omega_0^2 Q = \frac{F_0}{\omega} \sin \omega t$$

Complimentary Solution :

$$Q_c = e^{-\alpha t}(R_1 \cos \omega_d t + R_2 \sin \omega_d t)$$

$$\text{where } \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

Particular Solution :

$$Q_p = S_1 \cos \omega t + S_2 \sin \omega t$$

where

$$D_2 = (\omega_0^2 - \omega^2)^2 + (2\alpha\omega)^2$$

$$S_1 = -\frac{F_0(2\alpha\omega)}{D_2}$$

$$S_2 = \frac{F_0(\omega_0^2 - \omega^2)}{D_2\omega}$$

$$\therefore Q = e^{-\alpha t} \left[S_1 \cos \omega_d t + \left(\frac{\alpha S_1 - \omega S_2}{\omega_d} \right) \sin \omega_d t \right] + S_1 \cos \omega t + S_2 \sin \omega t$$